

MAXIMILIANO RODRIGUEZ

Principal Software Architect & Data Engineer

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EXECUTIVE SUMMARY

System Architect and Software Engineer with specialized focus in Zero-Trust cloud infrastructure (OCI), distributed event-driven pipelines, and high-performance AI inference engines. Pursuing a Bachelor of Science in Cyberdefense (UNDEF). Proven track record of architecting decoupled microservices in Spring Boot and FastAPI, reducing memory consumption by 82% via INT8 quantization and designing tamper-proof cryptographic audit ledgers.

PRODUCTION ENGINEERING PROJECTS

TECHMIND — Enterprise AI Microservices Platform

Remote

Principal Software Architect & Lead Engineer

Java 17, Spring Boot 3, FastAPI, ONNX, MySQL, OCI

- Optimized AI inference RAM from 2.5 GB to 110 MB (-82%) and latency from 1500ms to <18ms using INT8 quantization with ONNX Runtime and NumPy vectorization.
- Enforced 100% data deduplication and referential integrity across MySQL tables via content-based SHA-256 cryptographic hashing pipelines.
- Secured microservices with zero public IP exposure by architecting an isolated OCI Virtual Cloud Network (VCN) with NAT Gateway routing.

DATA SENTINEL — Zero-Trust Financial Compliance Ledger

Remote

Principal Software Architect

Python, FastAPI, PostgreSQL, AES-256-GCM, Scikit-Learn, SHAP

- Maintained sub-3ms API latency during CPU-bound ML inference by offloading Isolation Forest and SHAP models to an asynchronous ProcessPoolExecutor.
- Engineered an append-only cryptographic ledger with PL/pgSQL triggers, achieving 100% rejection of unauthorized UPDATE/DELETE attempts.
- Protected sensitive financial PII using AES-256-GCM authenticated encryption with key-rotation architecture.

AEGIS STREAM — Real-Time SIEM Telemetry Pipeline

Remote

System Architect & Data Engineer

ClickHouse, Redpanda, aiokafka, Docker, AsyncIO

- Handled peak telemetry ingestion of 50,000 events/second without event loop starvation via asynchronous backpressure controls.
- Eliminated duplicate event records during compaction phases using ClickHouse ReplacingMergeTree engine and Materialized Views.

TECHNICAL SKILLS

Cloud & Infrastructure: Oracle Cloud Infrastructure (OCI), Docker, Linux (Bash), NGINX, GitHub Actions, GitOps, CI/CD.
Languages & Runtimes: Python 3.12+, Java 17+, SQL / PL-SQL, JavaScript/TypeScript, HTML5/CSS3.
Frameworks & Tools: FastAPI, Spring Boot 3, ONNX Runtime, Pydantic V2, Pandas, NumPy, Scikit-Learn.
Databases & Streaming: PostgreSQL, ClickHouse, Redpanda (Kafka), MySQL 8.0, Redis, Oracle Autonomous Database.
Security & Standards: Zero-Trust Architecture, AES-256-GCM, Threat Intelligence, GRC Compliance, Cryptographic Ledgers.

EDUCATION & VERIFIED CREDENTIALS

Licenciatura en Ciberdefensa (Bachelor in Cyberdefense)

In Progress

Universidad de la Defensa Nacional (UNDEF)

Buenos Aires, Argentina

Oracle Cloud Infrastructure Certified Foundations Associate

Oracle

Verification ID: 103477615OCI26FNDCFA

Cloud Architecture & Security

Certificación Profesional en Ciberseguridad

IBM SkillsBuild

Network Security, Threat Intelligence & Active Defense

Credential ID Verified